

BUSINESS REQUIREMENTS DOCUMENT

EXPLORING SPACE, THE MILKY WAY GALAXY AND THE UNIVERSE

A Comprehensive Scientific Requirements, Knowledge and Exploration Framework

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Purpose of This Document

This Business Requirements Document (BRD) establishes a structured, scientifically grounded knowledge and requirements framework covering the origin, structure and evolution of the Universe, the Milky Way Galaxy, stellar and planetary systems, and the technical and research requirements associated with contemporary space exploration. It is intended to serve academic, research and professional documentation purposes.

Document Control

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Intended Audience

- Academic reviewers and students studying astronomy, astrophysics or planetary science
- Research teams preparing scientific project documentation or proposals
- Educational institutions developing space-science curricula
- Professionals producing structured technical documentation for outreach or portfolio use

Document Owner and Change-Control Process

The Office of Scientific Research Documentation (OSRD) is the designated owner of this document and is responsible for maintaining its accuracy over time. Any proposed change to factual content, terminology

or structure must be submitted as a change request, reviewed against current peer-reviewed scientific literature and data from recognized space agencies, and approved by the Lead Research Analyst before incorporation. Updates that reflect new observational data (for example, revised estimates of galactic mass or newly confirmed exoplanets) will be logged in the Version History table with a corresponding change description.

Document Outline (Parts 1–4)

This 40-page Business Requirements Document is delivered as four sequential parts (Part 1–4), each covering ten sections, so it can be reviewed or shared in smaller files. The full set should be read in order for a complete treatment of the subject.

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1. Executive Summary

The Universe encompasses all matter, energy, space and time, spanning scales from subatomic particles to galaxy superclusters distributed across tens of billions of light-years. This document presents a structured scientific and educational framework describing the origin, structure, contents and ongoing exploration of the cosmos, with particular emphasis on the Milky Way Galaxy, the stellar and planetary systems within it, and the scientific and technical requirements associated with studying them.

This Business Requirements Document (BRD) is organized to follow a logical progression: it begins with the large-scale origin and structure of the Universe, narrows focus to galaxies in general and the Milky Way in particular, proceeds through stellar formation, evolution and stellar remnants, addresses the Solar System and the search for life beyond Earth, and concludes with the technological, observational and research requirements that underpin modern space science.

Why Understanding the Cosmos Matters

- Cosmology and astrophysics test the fundamental laws of physics under conditions unreachable on Earth
- Studying stellar and planetary formation clarifies the origin of the chemical elements that constitute life
- Space-based observation underpins critical technologies including navigation, communication and Earth-monitoring systems
- Exploration of exoplanets and astrobiology informs humanity's understanding of its place in the Universe
- Long-term planetary and stellar risk assessment (e.g., solar activity, near-Earth objects) has direct relevance to planetary security

Major Scientific Areas Covered

- Cosmic origin, expansion and large-scale structure
- Galaxy classification, formation and the structure of the Milky Way
- Stellar life cycles, stellar remnants and high-energy astrophysical phenomena
- Dark matter, dark energy and the Cosmic Microwave Background

- The Solar System, exoplanets and astrobiology
- Space exploration technology, missions, observatories and future research directions

Key Open Questions Addressed

- What is the physical nature of dark matter and dark energy?
- What precisely triggered cosmic inflation, and what preceded the earliest observable epoch?
- Does life exist, or has it ever existed, elsewhere in the Solar System or beyond?
- What is the exact value of the Hubble constant, given discrepancies between measurement methods (the 'Hubble tension')?

Overall, this document is intended to function both as an educational reference and as a structured requirements framework that can guide future research planning, curriculum development or scientific project documentation. All content is drawn from established observational science, with theoretical and hypothetical material clearly distinguished from confirmed findings.

2. Introduction to the Universe

2.1 Definition of the Universe

The Universe is defined as the totality of space, time, matter and energy, together with the physical laws that govern their interaction. It contains all galaxies, stars, planets, interstellar and intergalactic matter, and the radiation that permeates space. Contemporary cosmology describes the Universe using the framework of general relativity, in which space and time are unified into a single four-dimensional structure known as spacetime.

2.2 The Observable Universe

The Observable Universe refers to the region of the cosmos from which light has had time to reach Earth since the Big Bang, approximately 13.8 billion years ago. Because space itself has continued to expand while that light travelled, the current radius of the Observable Universe is estimated at approximately 46.5 billion light-years, giving a diameter of roughly 93 billion light-years. This is a boundary of observation, not necessarily the physical edge of the Universe as a whole, which may be far larger or, according to some models, infinite.

2.3 Space and Time

Under general relativity, mass and energy curve the fabric of spacetime, and this curvature determines the motion of objects, which we perceive as gravity. Time is not absolute; it passes at different rates depending on relative velocity and gravitational field strength, an effect confirmed experimentally through atomic clock comparisons and satellite-based navigation systems, which must correct for relativistic time dilation to maintain accuracy.

2.4 Matter and Energy Content

Precision measurements of the Cosmic Microwave Background, most notably by the ESA Planck mission, indicate that the energy content of the Universe is distributed approximately as follows: ordinary (baryonic) matter accounts for about 5%, dark matter for about 27%, and dark energy for about 68%. This composition implies that the matter directly observable through electromagnetic radiation constitutes only a small fraction of the total energy budget of the cosmos.

Scale Reference Table

Quantity	Approximate Value	Notes
Age of the Universe	13.8 billion years	Derived from CMB and expansion measurements
Radius of Observable Universe	~46.5 billion light-years	Limited by light travel time and expansion
Ordinary (baryonic) matter	~5%	Stars, planets, gas, dust
Dark matter	~27%	Inferred gravitationally; composition unknown
Dark energy	~68%	Drives accelerated expansion
Estimated galaxy count (observable)	hundreds of billions	Order-of-magnitude estimate; revised with deeper surveys

Figure 1 – Conceptual diagram of the Observable Universe as a sphere centered on Earth, showing concentric shells representing increasing distance and lookback time, from nearby galaxies to the Cosmic Microwave Background at the outer boundary.

3. Origin of the Universe

3.1 The Big Bang Model

The Big Bang model is the prevailing scientific explanation for the origin and evolution of the Universe. It describes a state, approximately 13.8 billion years ago, in which the observable Universe was extremely hot, dense and rapidly expanding. The model does not describe an explosion into pre-existing space; rather, it describes the expansion of space itself from an initial hot, dense state.

3.2 Cosmic Inflation

Cosmic inflation is a theoretical extension to the standard Big Bang model, proposing an extremely rapid exponential expansion of space during a fraction of a second after the initial singularity-like state (roughly 10^{-36} to 10^{-32} seconds). Inflation is supported indirectly by the observed uniformity and flatness of the Universe and by the statistical pattern of tiny temperature fluctuations in the Cosmic Microwave Background, though it remains an area of active theoretical and observational research rather than a directly observed event.

3.3 Formation of Elementary Particles

As the early Universe cooled from its initial extreme temperatures, fundamental particles condensed out of the primordial energy field in a sequence governed by the Standard Model of particle physics: quarks and gluons formed first, subsequently combining into protons and neutrons (baryogenesis) within the first microsecond, followed by the formation of stable atomic nuclei.

3.4 Big Bang Nucleosynthesis

Between approximately 10 seconds and 20 minutes after the Big Bang, the Universe was hot and dense enough to fuse protons and neutrons into light atomic nuclei, chiefly hydrogen, helium-4, and trace amounts of deuterium and lithium-7. This process, called Big Bang nucleosynthesis, produced the

primordial elemental abundances still observed today in the oldest, most metal-poor stars and in intergalactic gas clouds, and its predicted ratios closely match observations, forming one of the strongest lines of evidence for the Big Bang model.

3.5 Evidence Supporting the Model

- Established evidence: the observed abundance ratios of light elements match Big Bang nucleosynthesis predictions
- Established evidence: the Cosmic Microwave Background radiation, discovered in 1965, matches the predicted afterglow of a hot early Universe
- Established evidence: the observed redshift of distant galaxies is consistent with an expanding Universe (Hubble-Lemaître law)
- Theoretical/unconfirmed: the precise mechanism and duration of cosmic inflation
- Open question: the physical state of the Universe prior to the Planck epoch is not currently describable by known physics

It is important to distinguish the well-established expansion history of the Universe, supported by multiple independent lines of observational evidence, from more speculative theoretical proposals — such as multiverse hypotheses or specific inflationary field models — which remain subjects of ongoing scientific investigation and are not yet confirmed by direct observation.

4. Cosmic Timeline

The evolution of the Universe from its earliest moments to the present day can be organized into distinct epochs, each characterized by specific physical conditions and processes. The table below summarizes this chronology, using approximate values drawn from standard cosmological models.

Epoch	Approximate Time	Key Characteristics
Planck Epoch	0 – 10^{-43} s	Known physics breaks down; quantum gravity effects dominate
Inflationary Epoch	10^{-36} – 10^{-32} s	Hypothesized exponential expansion of space
Quark Epoch	10^{-12} – 10^{-6} s	Universe filled with a quark-gluon plasma
Hadron Epoch	10^{-6} s – 1 s	Quarks combine into protons and neutrons
Nucleosynthesis	~10 s – 20 min	Light elements (H, He, Li) form
Recombination	~380,000 years	Neutral atoms form; photons decouple from matter (CMB released)
Cosmic Dark Ages	380,000 – ~200 million years	No luminous sources; neutral hydrogen fills space
First Stars (Population III)	~200 – 400 million years	First luminous objects ignite via gravitational collapse
First Galaxies	~400 million – 1 billion years	Protogalaxies assemble from gas and dark matter
Galaxy Evolution & Assembly	1 – 10 billion years	Galaxy mergers, star formation peaks, heavy

Epoch	Approximate Time	Key Characteristics
		elements accumulate
Solar System Formation	~9.2 billion years (4.6 Gya)	Sun and planets form from a collapsing molecular cloud
Present Universe	13.8 billion years	Accelerated expansion driven by dark energy

Interpretive Notes

- Times for the earliest epochs (Planck, inflationary, quark) are derived from theoretical extrapolation of particle physics, not direct observation
- The Recombination epoch is directly observable through the Cosmic Microwave Background
- The 'Cosmic Dark Ages' is named for the absence of visible light sources, not a physical absence of matter
- Estimates of first-star and first-galaxy formation timing are refined regularly using deep-field observations such as those from the James Webb Space Telescope

Figure 2 – Horizontal timeline diagram plotting cosmic epochs on a logarithmic time axis from the Planck epoch to the present day, with icons representing key transitions (nucleosynthesis, recombination, first starlight, galaxy assembly).

5. Scale of the Cosmos

5.1 Astronomical Distance Units

Because cosmic distances vastly exceed everyday experience, astronomers use a hierarchy of specialized units, each suited to a different scale of measurement, from the Solar System to the observable Universe.

Unit	Definition	Typical Use
Kilometer (km)	1,000 meters	Planetary diameters, spacecraft altitude
Astronomical Unit (AU)	~149.6 million km (Earth–Sun distance)	Solar System distances
Light-year (ly)	~9.46 trillion km (distance light travels in one year)	Distances to nearby stars
Parsec (pc)	~3.26 light-years	Stellar and galactic distances (standard professional unit)
Kiloparsec (kpc)	1,000 parsecs	Structure within and near the Milky Way
Megaparsec (Mpc)	1 million parsecs	Distances between galaxies and clusters
Gigaparsec (Gpc)	1 billion parsecs	Large-scale structure of the observable Universe

5.2 Comparative Scale Table

Reference Distance	Approximate Value
Earth to Moon	384,400 km
Earth to Sun (1 AU)	149.6 million km
Diameter of the Solar System (to the Oort Cloud)	~1–2 light-years
Distance to nearest star (Proxima Centauri)	~4.24 light-years
Diameter of the Milky Way disk	~100,000 light-years (estimates range up to ~150,000–200,000 ly for the full stellar disk)
Distance to the Andromeda Galaxy (M31)	~2.5 million light-years
Diameter of the Observable Universe	~93 billion light-years

These comparisons illustrate the extraordinary range of scales encompassed by astronomy: the Earth–Moon distance could be traversed by light in about 1.3 seconds, whereas light from the edge of the Observable Universe has been travelling, and space itself continuing to expand, for essentially the entire age of the cosmos.

Figure 3 – Logarithmic scale bar illustrating distance markers from Earth–Moon through Solar System, nearest stars, Milky Way diameter, nearest galaxies, and the Observable Universe.

6. Galaxies

6.1 Definition

A galaxy is a gravitationally bound system consisting of stars, stellar remnants, interstellar gas and dust, and dark matter, typically containing anywhere from a few hundred million to several trillion stars. Galaxies are the primary structural units in which star formation and stellar evolution occur across cosmic history.

6.2 Morphological Classification

Galaxies are commonly classified using a scheme originally proposed by Edwin Hubble, which groups them by visual structure into spiral, elliptical and irregular forms, later extended to include lenticular and dwarf categories.

Type	Description	Example
Spiral	Flattened rotating disk with spiral arms, a central bulge, and ongoing star formation	Milky Way, Andromeda (M31)
Barred Spiral	Spiral galaxy with a central elongated bar structure feeding the arms	Milky Way (barred spiral)
Elliptical	Smooth, ellipsoidal distribution of older stars with little gas and low star-formation activity	M87
Lenticular	Disk galaxy without prominent spiral arms; intermediate between spiral and elliptical	NGC 5866

Type	Description	Example
Irregular	Lacks a defined symmetric structure, often due to gravitational interaction	Large Magellanic Cloud
Dwarf	Small galaxies (fewer stars) that may be spiral, elliptical or irregular in form	Sagittarius Dwarf, Small Magellanic Cloud

6.3 Galaxy Clusters and Mergers

Galaxies rarely exist in isolation. Most belong to groups or clusters bound by mutual gravity, and interactions between galaxies — ranging from tidal distortion to full mergers — play a central role in galaxy evolution. Merger events can trigger intense bursts of star formation, redistribute gas and dust, and, over long timescales, transform spiral galaxies into elliptical remnants.

6.4 Galaxy Evolution

Galaxy evolution refers to the long-term changes in a galaxy's structure, stellar population and star-formation activity over cosmic time, driven by internal processes (star formation, stellar feedback, central black hole activity) and external processes (mergers, gas accretion, environmental stripping in clusters). Observations of galaxies at different distances — and therefore different look-back times — allow astronomers to reconstruct this evolutionary history directly.

Figure 4 – The Hubble morphological classification 'tuning fork' diagram, showing elliptical galaxies branching into barred and unbarred spiral sequences.

7. The Milky Way Galaxy — Overview

The Milky Way is the barred spiral galaxy that contains the Solar System. It is a member of the Local Group, a small collection of more than 80 galaxies dominated by the Milky Way and the Andromeda Galaxy. Direct imaging of the Milky Way's overall structure from outside is not possible, since Earth is located within its disk; its large-scale structure is therefore inferred from radio, infrared and stellar-motion surveys combined with analogy to similar external spiral galaxies.

7.1 Key Structural Parameters

Property	Approximate Value	Note
Type	Barred spiral galaxy (SBbc)	Classification refined via infrared surveys
Stellar disk diameter	~100,000–200,000 light-years	Estimates vary with the faint outer disk definition
Disk thickness (thin disk)	~1,000 light-years	Thicker disk component extends further
Number of stars	~100–400 billion	Order-of-magnitude estimate
Total mass (including dark matter)	~1–1.5 trillion solar masses	Dominated by the dark matter halo
Sun's distance from Galactic	~26,000–27,000 light-years	Refined via stellar orbit

Property	Approximate Value	Note
Center		studies
Sun's orbital period around the Galactic Center	~225–250 million years ('galactic year')	

7.2 Major Structural Components

- Galactic disk — contains the spiral arms, most young stars, and the interstellar medium
- Central bulge — a dense, roughly spheroidal concentration of older stars surrounding the galactic center
- Stellar halo — a sparse, roughly spherical distribution of old stars and globular clusters surrounding the disk
- Dark matter halo — an extended, gravitationally dominant component inferred from the galaxy's rotation curve

7.3 Location of the Solar System

The Solar System resides within a minor spiral structure known as the Orion Arm (or Orion Spur), located between the larger Sagittarius and Perseus arms, roughly two-thirds of the way from the Galactic Center to the visible edge of the disk. This position places the Sun within the thin disk, in a region of moderate stellar density relative to the crowded galactic bulge.

Figure 5 – Face-on schematic of the Milky Way showing the central bar, major spiral arms (Perseus, Sagittarius, Scutum-Centaurus, Norma) and the Sun's position within the Orion Arm.

8. Detailed Structure of the Milky Way

8.1 The Galactic Disk

The galactic disk is subdivided into a thin disk and a thick disk. The thin disk, roughly 1,000 light-years thick, hosts most young, metal-rich stars and active star-forming regions, and contains the visible spiral arms. The thick disk extends several thousand light-years above and below the plane and is populated by somewhat older, more metal-poor stars, suggesting it formed earlier in the galaxy's history or was thickened by past merger events.

8.2 The Central Bulge

The bulge is a dense, elongated (boxy/peanut-shaped) concentration of stars surrounding the galactic center, extending roughly 10,000–16,000 light-years in radius. Infrared surveys, which penetrate the obscuring dust of the disk, reveal that the bulge is dominated by old, metal-rich stars and contains a pronounced bar structure that channels gas toward the galactic center.

8.3 The Stellar Halo and Globular Clusters

The stellar halo is a diffuse, roughly spherical distribution of old, metal-poor stars extending far beyond the disk. It hosts the majority of the Milky Way's approximately 150 known globular clusters — dense, gravitationally bound spherical collections of up to a million ancient stars — some of which are among the oldest known objects in the galaxy, providing constraints on early galactic formation.

8.4 Spiral Arms

The Milky Way's spiral arms are regions of enhanced gas density and star formation, traced observationally through the positions of young massive stars, ionized hydrogen (H II) regions, and molecular clouds. Major named arms include the Perseus Arm, Sagittarius Arm, Scutum-Centaurus Arm and Norma Arm, connected to the central bar.

8.5 The Interstellar Medium

The interstellar medium (ISM) consists of the gas and dust occupying the space between stars within the disk, comprising roughly 10–15% of the disk's visible mass. It includes diffuse atomic and molecular hydrogen clouds, ionized regions surrounding hot stars, and dust grains that absorb and redden starlight — the raw material from which new stars continually form.

Figure 6 – Cross-sectional (edge-on) diagram of the Milky Way showing the thin disk, thick disk, central bulge, and surrounding spherical stellar halo with globular clusters.

8.6 Galactic Rotation

Unlike a rigid body, the Milky Way exhibits differential rotation: stars at different galactocentric radii orbit at different angular rates. Measured rotation curves — orbital velocity as a function of radius — remain roughly flat at large radii rather than declining as expected from the visible mass distribution alone, a key piece of evidence for the presence of a substantial dark matter halo.

9. The Galactic Center and Sagittarius A*

9.1 Overview of the Galactic Center Region

The Galactic Center lies in the direction of the constellation Sagittarius and is heavily obscured at visible wavelengths by intervening dust and gas in the disk. Consequently, it is studied primarily through infrared, radio and X-ray observations, which penetrate this obscuring material.

9.2 Sagittarius A*

At the dynamical center of the Milky Way lies Sagittarius A* (Sgr A*), a compact, extremely massive radio source identified as a supermassive black hole. Long-term tracking of the orbits of individual stars closely surrounding Sgr A* — work recognized by the 2020 Nobel Prize in Physics — has determined its mass to be approximately 4.3 million times the mass of the Sun, concentrated within a region smaller than the orbit of Mercury around the Sun.

9.3 Direct Imaging

In 2022, the Event Horizon Telescope collaboration released the first direct image of the shadow cast by Sgr A* against its surrounding emission, following a similar image of the black hole in the galaxy M87 released in 2019. These images provide direct visual confirmation consistent with general relativity's predictions for a black hole's event horizon.

9.4 Stellar Orbits and Accretion

- Individual stars (the 'S-stars', notably S2) have been tracked through complete orbits around Sgr A*, providing precise mass measurements
- Sgr A* is comparatively faint and quiescent relative to active galactic nuclei in other galaxies, indicating a low current rate of matter accretion

- Occasional flares in X-ray and infrared emission are attributed to sporadic accretion of gas or dust onto the black hole
- A dense cluster of stars and stellar remnants surrounds Sgr A*, complicating direct observation of the black hole itself

9.5 Observational Challenges

Studying the Galactic Center presents several persistent challenges: interstellar dust extinguishes visible light by many orders of magnitude along this line of sight; the region is crowded with foreground stars that complicate isolating individual sources; and the black hole's small angular size on the sky (despite its enormous mass) requires the extremely high angular resolution achieved only by combining multiple telescopes across the globe, as in the Event Horizon Telescope array.

Figure 7 – Composite infrared/X-ray/radio image style illustration of the Galactic Center region, highlighting the orbits of the S-stars around Sagittarius A.*

10. Stars

10.1 What Is a Star?

A star is a massive, self-luminous sphere of plasma held together by its own gravity, generating energy primarily through nuclear fusion in its core. The overwhelming majority of stars fuse hydrogen into helium, a process that releases enormous quantities of energy in accordance with Einstein's mass-energy relation.

10.2 Stellar Composition and Nuclear Fusion

Stars are composed predominantly of hydrogen (roughly 70–75% by mass) and helium (roughly 25%), with trace amounts of heavier elements referred to collectively by astronomers as 'metals.' In the core of a main-sequence star, extreme temperature and pressure enable the proton-proton chain (in lower-mass stars like the Sun) or the CNO cycle (in more massive, hotter stars) to fuse hydrogen nuclei into helium, releasing the energy that ultimately emerges as starlight.

10.3 Mass, Luminosity and Temperature

A star's mass is the single most important factor determining its luminosity, surface temperature, color, lifespan and ultimate fate. More massive stars burn their nuclear fuel more rapidly and thus have shorter lifespans despite possessing larger fuel reserves, following an approximate mass-luminosity relationship in which luminosity scales roughly with mass to a power between 3 and 4.

10.4 Spectral Classification

Stars are classified by spectral type — a sequence historically labeled O, B, A, F, G, K, M, from hottest to coolest — based on the absorption features in their spectra, which correlate directly with surface temperature.

Spectral Class	Approx. Surface Temp. (K)	Color	Example
O	> 30,000	Blue	Zeta Puppis
B	10,000 – 30,000	Blue-white	Rigel
A	7,500 – 10,000	White	Sirius A

Spectral Class	Approx. Surface Temp. (K)	Color	Example
F	6,000 – 7,500	Yellow-white	Procyon A
G	5,200 – 6,000	Yellow	Sun
K	3,700 – 5,200	Orange	Arcturus
M	< 3,700	Red	Proxima Centauri

10.5 The Main Sequence